

Xiyue Zhu

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EDUCATION

- Aug 2021 – Dec 2026 **Ph.D. in Computer Engineering**, University of Illinois Urbana-Champaign
Expected Dec 2026 · Advisor: Prof. Volodymyr Kindratenko
Research focus: VLM/LLM Post-training (SFT, Distillation, DPO, GRPO) · Agentic Systems · Video & Diffusion Generation
GPA: 3.92/4.00
- Aug 2021 – May 2023 **M.S. in Computer Engineering**, University of Illinois Urbana-Champaign
- Aug 2017 – May 2021 **B.Eng. (Zhejiang University) & B.Sc. (UIUC) in Computer Engineering**, ZJU-UIUC Institute dual-degree program
GPA: 3.88/4.00 (UIUC), 3.87/4.00 (ZJU)

WORK EXPERIENCE

Machine Learning Engineer Intern, TikTok May 2026 – Aug 2026

- **Project:** Query Recommendation and Agentic Search for E-commerce Product Pages and Videos
- Post-trained Qwen3-VL-8B for query recommendation: distillation from a frontier closed-source model on 200K samples, then GRPO with rewards from online CTR of deployed questions and agent-as-judge.
- A downstream web agent answers proposed questions via tool-calling internal search.
- Trained with verl. Doubled offline-estimated CTR over the production system. (closed-source model)

Machine Learning Intern, Xpeng Motors May 2025 – Jan 2026

- **Project:** Unified BEV Space for VLM-based End-to-end Driving and Planning
- Designed a unified BEV representation with human readable BEV images for VLA driving model.
- This representation is shared by planning and end-to-end driving, enabling pretraining on large-scale vectorized planning data before fine-tuning on end-to-end driving.
- Led distributed training on 200+ GPUs; reduced L2 trajectory error by 4–9% over the baseline model.

Applied Scientist Intern, Amazon May 2024 – Aug 2024

- **Project:** Turbocharging Web Automation: The Impact of Compressed History States
- Designed a compressed history-state representation for VLM web agents, cutting context length by ~70% while preserving the action history needed for multi-step navigation.
- Improved step accuracy by 2–5% on Mind2Web and WebLINX; published at ACL 2025 Findings.

RESEARCH EXPERIENCE

Agentic Reasoning for Video Generation Jan 2025 – June 2026

- Proposed a video-creation agent that integrates chain-of-thought planning and iterative self-refinement into video generation, built on Causal Forcing as the generator and Qwen2.5-VL-7B as the reasoner.
- Trained the reasoner with GRPO using VisionReward on the final video as the reward signal, improving VisionReward by 20.3% over the base generator, with better physical plausibility and causal consistency.
- Supports multi-modal conditioned video generation through multi-modal understanding.
- Enables interactive, interpretable generation through explicit, editable reasoning traces.

Medical Question Proposing and Answering for Report Generation

Jun 2025 – Jan 2026

- Proposed **MedQPA**, a question-driven agentic evaluation pipeline for medical report generation.
- Developed **MedQPA-Gen**, which optimizes report generators against MedQPA via reflective prompting, iterative DPO, and GRPO on Qwen2-VL-7B.
- Achieved an 80% human-preference win rate over the base model and +3.5% accuracy on downstream disease classification; ACL 2026 Findings (co-first author).

Score Fusion in Diffusion Models

Nov 2023 – Feb 2025

- Proposed **Score Fusion**, which learns to fuse diffusion score functions instead of combining them arithmetically, yielding a 3D representation for volume-to-volume translation and multi-modal fusion.
- Improved 3D medical-image and video inverse-problem results by 3–10%; ICML 2025.

MapPrior: BEV Map Layout Estimation with Generative Models

Jul 2022 – Aug 2023

- Introduced generative priors for BEV map segmentation, exploiting the prior distribution of map layouts.
- Improved IoU by $\sim 1\%$ and substantially improved realism and uncertainty estimation for downstream planning; ICCV 2023.

LiDARGen: Generate Realistic LiDAR Point Clouds

Jan 2022 – Oct 2022

- Implemented BEV-FID metrics and LiDAR generative baselines (ShapeGF, PU-GAN); ECCV 2022.

PUBLICATIONS

- [1] W. Liang*, **X. Zhu**[†], R. Zhu, C. Li, C. Tang, Z. Liu, Z. Gong, S. Luo, Y. Li, V. Kindratenko. *MedQPA-Gen: Medical Question Proposing and Answering for Report Generation*. ACL 2026 Findings.
- [2] **X. Zhu**, D.H. Kwark, R. Zhu, K. Hong, Y. Tao, S. Luo, Y. Li, Z.P. Liang, V. Kindratenko. *Introducing 3D Representation for Volume-to-Volume Translation via Score Fusion*. ICML 2025.
- [3] **X. Zhu**, P. Tang, H. Liao, S. Appalaraju. *Turbocharging Web Automation: The Impact of Compressed History States*. ACL 2025 Findings.
- [4] **X. Zhu**, V. Zyrianov, Z. Liu, S. Wang. *MapPrior: Bird’s-Eye View Map Layout Estimation with Generative Models*. ICCV 2023.
- [5] V. Zyrianov, **X. Zhu**, S. Wang. *Learning to Generate Realistic LiDAR Point Clouds*. ECCV 2022.
- [6] A. Saxton, J. Dong, A. Bode, N. Jaroenchai, R. Kooper, **X. Zhu**, et al. *Accurate Feature Extraction from Historical Geologic Maps Using Open-Set Segmentation and Detection*. Geosciences 2024.

* Equal contribution † Corresponding author

ACADEMIC SERVICE

Reviewer: CVPR, ECCV, NeurIPS

SKILLS

Frameworks PyTorch, FSDP, verl, vLLM, SGLang

Areas LLM/VLM post-training (SFT, distillation, iterative DPO, GRPO), reward design from on-line feedback and LLM-as-judge, tool-using agents, diffusion & video generation, distributed training at 200+ GPU scale

Languages Python, C++, CUDA